SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Pseudocode Writing Task (Algorithm Design)

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO5 – Articulation	**Assessment:**	Assessment Tool 2 – Pseudocode Writing Task (Algorithm Design)
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
Student Name:	K.Mounith	**Reg. No.:**	192425090
Section / Batch:	AIML/2024	**Date:**	18-08-2026

Code of Conduct

I, K.Mounith(192425090)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K.Mounith

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Reference Resolution, Discourse	Entity Linking Algorithm	Q1
Text Coherence, Discourse Structure	Coherence Analysis Algorithm	Q2
Dialog Acts, Conversational Agents	Intent Recognition Algorithm	Q3
Language Generation, Surface Realization	NLG Pipeline Design	Q4
Machine Translation, Interlingua, Statistical Approaches	Translation Workflow Algorithm	Q5

Annexure A – Pseudocode Writing Task (Algorithm Design)

1. Design an algorithm to perform **Reference Resolution** by identifying pronouns and linking them to the correct entities in a discourse. Apply your algorithm to the text:

"Ravi met Arun at the library. He borrowed a book and later returned it."

Generate the resolved discourse by replacing pronouns with their corresponding entities. Further, validate the correctness of the resolution process by explaining how discourse context is used to identify the antecedents.

A. ALGORITHM Reference_Resolution

```
INPUT:
   Discourse = "Ravi met Arun at the library.
               He borrowed a book and later returned it."
OUTPUT:
   Resolved Discourse
BEGIN
   Identify all entities:
      Ravi, Arun, library, book
   Identify pronouns:
      He, it
   FOR each pronoun DO
      Find possible antecedents from previous text
      IF pronoun = "He" THEN
         Check gender and number
         Check discourse context
         Select "Ravi" as antecedent
      END IF
      IF pronoun = "it" THEN
         Check singular object
         Check semantic context
         Select "book" as antecedent
      END IF
      Replace pronoun with selected antecedent
```

END FOR

Display Resolved Discourse

END

2. Develop an algorithm to analyze **Text Coherence** and **Discourse Structure** by identifying logical relationships between sentences such as cause-effect, sequence, and elaboration. Apply your algorithm to the following discourse:

"The roads were flooded after heavy rainfall. Therefore, schools were closed for the day. Students attended classes online."

Identify the discourse relations among the sentences and construct a discourse structure representation. Further, justify how these relations contribute to maintaining coherence in the text.

A. ALGORITHM Discourse_Analysis

```
INPUT:
   Three sentences S1, S2, S3

OUTPUT:
   Discourse relations and coherence

BEGIN

   S1 ← "The roads were flooded after heavy rainfall."
   S2 ← "Therefore, schools were closed for the day."
   S3 ← "Students attended classes online."

   Identify the main events in each sentence

   IF S2 is caused by S1 THEN
      Relation(S1,S2) ← CAUSE-EFFECT
   END IF

   IF S3 happens after S2 THEN
      Relation(S2,S3) ← SEQUENCE
   END IF

   Construct discourse structure

   Check whether all sentences are logically connected

   IF connected THEN
      Coherence ← HIGH
   ELSE
      Coherence ← LOW
   END IF

   Display relations and structure
```

END

3. Design an algorithm for a **Conversational Agent** that identifies **Dialogue Acts** from user interactions. Apply your algorithm to the conversation:

User: "Can you book a train ticket for me?"
Agent: "Sure, where would you like to travel?"
User: "I want to go to Chennai."
Agent: "Your ticket has been booked."

Classify each utterance as Request, Question, Inform, Confirmation, or Action. Generate the dialogueact sequence and evaluate how the identified acts help the conversational agent interpret user intentions.

A.ALGORITHM Dialogue_Act_Identification

INPUT:

Conversation utterances

OUTPUT:

Dialogue Act for each utterance

BEGIN

FOR each utterance DO

IF user asks for a service THEN

Act ← REQUEST

ELSE IF utterance asks for information THEN

Act ← QUESTION

ELSE IF user provides information THEN

```
      Act ← INFORM

    ELSE IF utterance confirms something THEN
      Act ← CONFIRMATION

    ELSE IF agent performs the requested task THEN
      Act ← ACTION

    END IF

    Display utterance and Act

  END FOR

END
```

User: "Can you book a train ticket for me?"

↓

REQUEST

↓

Agent: "Sure, where would you like to travel?"

↓

QUESTION

↓

User: "I want to go to Chennai."

↓

INFORM

↓

Agent: "Your ticket has been booked."

↓

ACTION

4. Design an algorithm for **Language Generation** that converts a semantic representation into a grammatically correct sentence through **Surface Realization**. Consider the semantic input:

```
Action: Buy
Agent: Student
Object: Book
Tense: Past
```

Generate the final natural language sentence and explain how the algorithm performs lexical selection, sentence structuring, and surface realization. Further, validate the grammatical correctness of the generated output.

Language Generation – Surface Realization

a) Algorithm / Pseudocode

ALGORITHM Language_Generation

INPUT:

Action = Buy

Agent = Student

Object = Book

Tense = Past

BEGIN

Lexical Selection:

Agent → "The student"

Action → "buy"

Object → "a book"

Apply Tense:

"buy" → "bought"

Sentence Structuring:

Arrange as Agent + Verb + Object

Surface Realization:

Generate complete sentence

OUTPUT:

"The student bought a book."

END

b) Semantic Structure

```
          BUY
         /  \
        /    \
   STUDENT   BOOK
    Agent    Object
         |
      Past Tense
```

c) Surface Realization

Semantic Input

↓

Lexical Selection

↓

Student + Buy + Book

↓

Past Tense

↓

Bought

↓

Sentence Structure

↓

Agent + Verb + Object

↓

"The student bought a book."

d) Explanation

Lexical Selection: Selects suitable words: *student, buy, book*.

Sentence Structuring: Arranges them using the Subject + Verb + Object (SVO) structure.

Surface Realization: Changes *buy* to bought to express past tense and adds appropriate articles.

e) Validation

Subject → The student ✓

Verb → bought ✓ Past tense

Object → a book ✓

Word Order → S + V + O ✓

Grammar → Correct ✓

Final Natural Language Sentence:
“The student bought a book.”

5. Design an algorithm that combines **Interlingua-based Machine Translation** and **Statistical Translation Approaches**. Apply your algorithm to translate the sentence:

"The boy is playing football."

Perform the following steps:

1. Analyze the source sentence.
2. Create an Interlingua representation.

3. Generate candidate target-language translations.
4. Apply statistical scoring to select the most appropriate translation.
5. Produce the final translated sentence.

Finally, evaluate how the Interlingua representation and statistical methods improve translation quality and reduce ambiguity.

A. Interlingua + Statistical Machine Translation

Algorithm / Pseudocode

```
ALGORITHM Hybrid_Machine_Translation

INPUT:

    Source = "The boy is playing football."

OUTPUT:

    Target Translation

BEGIN

    Analyze source sentence

    Identify:

        Subject = boy

        Action = play

        Object = football

        Tense = Present Continuous

    Create Interlingua:

        AGENT(boy)

        ACTION(play)

        OBJECT(football)

        TENSE(present_continuous)

    Generate candidate translations:

        Candidate 1 = "The boy is playing football."

        Candidate 2 = "The boy plays football."

        Candidate 3 = "The boy is playing the football."
```

Calculate statistical scores using:

Language Model Score

Translation Probability

Grammar Score

Select candidate with highest score

Produce final translation

END

1. Source Sentence Analysis

```
The boy is playing football.

  |      |      |

 Subject  Action  Object

  |      |      |

  Boy     Play   Football
```

Tense: Present Continuous

2. Interlingua Representation

```
      PLAY

     /   \

    ↓     ↓

  BOY    FOOTBALL

 Agent    Object

    |

    ↓

Present Continuous
```

AGENT(boy)

ACTION(play)

OBJECT(football)

3. Candidate Target Sentences

C1 → The boy is playing football.

C2 → The boy plays football.

C3 → The boy is playing the football.

4. Statistical Scoring

Candidate	Score
C1: The boy is playing football	0.95 ✓
C2: The boy plays football	0.70
C3: The boy is playing the football	0.60

Highest score → C1

5. Final Translation

The boy is playing football.

Evaluation

Source Sentence

↓

Interlingua

↓

Meaning Preservation

↓

Candidate Generation

↓

Statistical Scoring

↓

Best Translation

- Interlingua preserves the core meaning: boy + play + football + present continuous.
- Statistical scoring selects the most natural grammatical candidate.

- The combination reduces ambiguity because Interlingua handles meaning, while the statistical model selects the most probable surface form.
- Thus, the hybrid approach provides better semantic accuracy, grammatical correctness, and natural language output.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

Q. No. / Criteria Level	Problem Understanding	Algorithm Design	Use of Control Structures	Pseudocode Structure	Completeness	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________